

K.K. Knitwear Club — Website Rebuild

Rebuilding a Ludhiana textile manufacturer's IndiaMART-hosted template into a searchable, filterable, comparison-driven B2B fabric catalogue. Front-end only, deployed to Vercel.

Live: kk-knitwear-club.vercel.app | Repo: github.com/SimranjitKaur543/kk-knitwear-club | Stack: vanilla HTML/CSS/JS, no framework, no build step | Verified at commit `e0c05c3`

1. STARTING POINT

The existing website was a hosted IndiaMART template. Its stylesheets and scripts were served from `tdw.imimg.com`, its product photographs from `5.imimg.com`, and it loaded jQuery 1.x alongside three Google Analytics tags. There was no source code to refactor.

That finding reframed the work: not an incremental redesign, but a rebuild that had to faithfully carry over content published on someone else's platform, without deleting anything useful.

2. OLD VERSUS NEW

Aspect	Old site	New site
Platform	IndiaMART template, external CSS/JS, jQuery 1.x	Own vanilla HTML/CSS/JS, no dependencies
Product pages	None — products were anchors on long category pages	One templated page driving all 50 products
Catalogue	48 listings, 17 categories (9 held one product)	50 products, 8 end-use categories
Search	None	Live search across names, trade terms and all specs
Filtering	None	9 groups, 58 options (GSM, width, construction, etc.)
Comparison	None	Up to 4 fabrics side by side, differences highlighted
Shortlist	None	localStorage, persists, feeds a bulk enquiry
Enquiry	"Send Email" / "Send SMS" links only	Product-prefilled form, validation, WhatsApp/email handoff
Images	Hotlinked at 250x250	96 self-hosted, most 1000px on the long edge
Navigation	3 items, no breadcrumbs	4 items, breadcrumbs on deep pages, mobile drawer
Deployment	Third-party hosting	GitHub to Vercel, custom 404, noindex headers

Note: product count rose from 48 to 50 because the Polyester Fabric page carried 21 listings, including two distinct entries sharing one anchor. Both were preserved.

3. MAJOR CHANGES AND FEATURES ADDED

- Two-layer specification model. The mill publishes 51 distinct spec keys across six alias families (GSM alone appears as "GSM", "Fabric GSM" and "Products GSM"). Published values are stored and displayed verbatim; a separate derived layer is computed at runtime purely to power search, filters and comparison.
- Taxonomy rebuilt around end use. 17 categories, 9 of them holding a single product with overlapping names, became 8 grouped by what the buyer is making. Every original category name is kept as a searchable tag.
- Complete buyer journey: Discover, Search, Filter, Understand, Compare, Enquire — including a shortlist that converts a whole saved list into one enquiry.
- Quick Specs and Best Suited For. GSM, width, composition, construction, pattern and finishing surface above the fold; applications are derived from the mill's own usage fields into a consistent set of 16.
- Generated short descriptions. Only 4 of 50 products publish a description; the other 46 receive one sentence composed by code from already-published values — a restatement, never invention.
- Customer reviews recovered. Eight buyer reviews from the original testimonial page, missed on the first pass and caught by a later self-audit, were restored verbatim.

4. WORKFLOW FOLLOWED

- Inspect and analyse. Audited the live DOM, established no local codebase existed, mapped 17 category pages and every listing. Produced a written analysis before proposing anything.
- Plan and approval gate. Agreed stack, image strategy, enquiry approach, spec-normalisation policy and deployment intent before writing code.
- Extraction. Scripted DOM extraction across all 17 category pages; found the CDN exposed 1000x1000 originals through a data-multiimg attribute and downloaded 96 images with resolution fallbacks.
- Data modelling. Designed the verbatim/derived split, deduplicated near-identical listings, validated coverage before building any UI.
- Build. Design system, catalogue with search and filters, templated product page, compare, shortlist, enquiry, static pages.
- UI and motion passes. Typography and spacing refinement, then scroll reveal, image fade-in and a word-by-word hero headline — each checked for reduced-motion and no-JavaScript behaviour.
- Requirements audit. Checked the build against every clause of the brief; two genuine gaps were found and closed.
- Deployment. Pre-flight checks, Git init, GitHub push, Vercel import, noindex headers and a custom 404.
- Colour system iterations. Three successive palettes, each measured for contrast before adoption.

5. APPROXIMATE TIME SPENT

Time was not instrumented during the project. Git records four commits spanning 21:28 to 22:29 on 27 August 2026, but that window covers only the committed portion — extraction, data modelling and the initial build all preceded the first commit. The distribution below is relative effort, inferred from code volume and the number of review cycles each phase required, not measured hours. Replace with your own recorded times if you kept a log.

Phase	Share of effort
Extraction and data modelling	~22%
Core build (catalogue, detail, compare, enquiry)	~30%

UI, motion and colour iterations	~24%
Testing, auditing and debugging	~16%
Deployment and documentation	~8%

6. TOOLS AND TECHNIQUES

Area	Tool / technique
Extraction	Browser DOM scripting — rendered markup exposed 1000x1000 image URLs the visible page never used
Bulk download	PowerShell with resolution fallback: try 1000x1000, fall back to 500x500, then 250x250
Data layer	Plain JavaScript object array plus a runtime normaliser — no database; one file to edit per product
Validation	Node scripts checking asset references, duplicate ids, orphan images and category coverage
Accessibility	WCAG contrast calculation, then auditing computed styles on the rendered pages
State	localStorage with an in-memory fallback — private browsing throws rather than returning empty
Motion	IntersectionObserver plus a throttled scroll sweep — observer is the fast path, geometry is the guarantee
Deployment	Git, GitHub, Vercel and vercel.json for header control and redeploy on push

7. BEYOND RAW AI PROMPTS

The build was not a single generation. The substantive effort was in specifying, verifying and correcting. Most defects below were found by testing, not by reading the code.

Verification tooling written alongside the site

- A validator checking all 163 asset references, unique product ids, orphaned images and empty categories.
- A deployment pre-flight catching case-sensitivity mismatches — Windows is case-insensitive, every host runs Linux — plus absolute paths and insecure URLs.
- A contrast calculator run before each palette was adopted, and a DOM auditor run after.

Defects found and fixed through testing

Defect	Root cause
Every section label rendered grey, not accent	.section-head p at specificity (0,1,1) silently beat .eyebrow at (0,1,0)
Hero collage ignored its aspect ratio	1fr tracks cannot shrink below content; needed minmax(0,1fr)
Scroll reveal would have switched itself off	A safety timer revealed all tagged elements, marking below-fold content as seen
Sections could stay permanently blank	IntersectionObserver callbacks ride the rendering loop; a never-painted document never receives them

Labelled Shortlist button collapsed to a bare icon	A shared sync routine overwrote innerHTML on every toggle, not only icon ones
Filter checkboxes lost their accent colour	A palette swap left var(--teal) referencing a deleted token
Card buttons overflowed by 6px on mobile	Three controls could not fit one row in a two-up grid at 360px
Push rejected by GitHub (GH007)	Commit authored with a private email; fixed with a noreply relay and --amend --reset-author

Judgement calls on data integrity

- One fabric publishes Width: 37 mm, implausible for cloth. It is still displayed as published but excluded from width filtering. The same rule covers "custom" and "on order": shown, not guessed.
- The source testimonial page contradicts itself — its star breakdown counts a one-star review it never displays. Averaging the eight visible reviews would have printed an inflated 5.0, so the mill's own published 4.5 is quoted instead.
- Five products shared names but had different specifications. All were kept, disambiguated by a published value rather than deleted as duplicates.

8. PROBLEMS AND HOW THEY WERE SOLVED

Problem	Resolution
No source code existed to modify	Reverse-engineered content from the rendered DOM; rebuilt with owned code
51 spec keys across 6 alias families, with values such as "120 to 250 GSM" and "on order"	Verbatim storage for display; a runtime normaliser for filtering that omits rather than guesses
Brand colours failed accessibility — every accent in the final palette fell below AA as text and as a fill with white text	Given hues kept for fills, borders and section grounds; deepened shades of the same hues carry all text
Four tinted section backgrounds were mutually indistinguishable (1.00–1.03 ratio)	Established a rule that the neutral ground must separate any two tinted sections
Preview tooling stopped rendering mid-project	Switched to auditing computed styles and geometry directly — stricter than screenshots for colour work
Deployment could break on a case-sensitive host	Wrote a pre-flight verifying all 163 references match on-disk casing exactly

9. KEY TECHNICAL LEARNINGS

- CSS specificity fails silently. A one-element difference between (0,1,1) and (0,1,0) removed an accent colour site-wide with no error anywhere.
- 1fr is not minmax(0,1fr). Grid tracks default to a min-content floor, so an image can force a track past its intended size.
- IntersectionObserver is not a guarantee. Its callbacks are delivered as part of the rendering loop, so anything hidden pending a callback needs a geometry-based fallback.
- Brand palettes rarely survive contrast maths. Mid-tone hues sit in a dead zone: too dark for white text, too light for a dark ground. Deepened variants of the same hue preserve the identity while restoring legibility.

- Windows hides deployment bugs. A case-insensitive filesystem lets broken references pass locally and fail on any Linux host.
- Progressive enhancement needs an explicit off-state. Anything hidden by JavaScript must be visible when that JavaScript does not run.

10. TRANSFERABLE LESSONS

- Measure before deciding. Two palettes were adjusted because the numbers said so, not because they looked wrong. Calculating first is faster than redesigning after a review.
- Build the verifier, not just the thing. The validator and pre-flight scripts caught more defects than reading the code did, and kept catching them on every later change.
- Separate source data from derived data. Storing published values verbatim and deriving everything else made it possible to add search, filters and generated descriptions without ever altering a source fact.
- Audit your own work against the brief. A deliberate pass against each requirement found two real gaps that normal testing would not have surfaced.
- A safety net can defeat the feature it protects. The scroll-reveal fallback would have quietly disabled scroll reveal; fallbacks need testing as carefully as the paths they guard.
- Report contradictions rather than resolving them silently. Where source data disagreed with itself, quoting it and flagging the conflict was more defensible than picking the flattering number.

KNOWN LIMITATIONS

This is a front-end demo. The enquiry form validates and composes a message but has no server behind it, and COMPANY.whatsapp still holds the original listing's call-routing number rather than a real mobile. Both are documented in the repository README. All figures in this document were verified against the repository at commit e0c05c3 and the live deployment; no product specification was invented.